DRAFT — NOT FOR FABRICATION. Functional architecture only; component selection is not frozen.

power_input



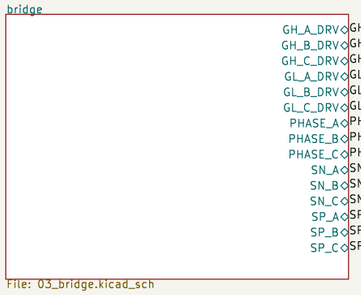
File: 01_power_input.kicad_sch

dc_link



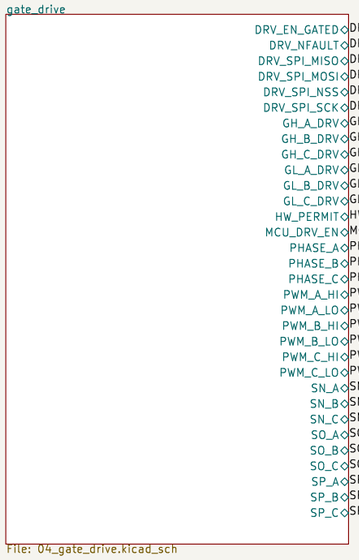
File: 02_dc_link.kicad_sch

bridge



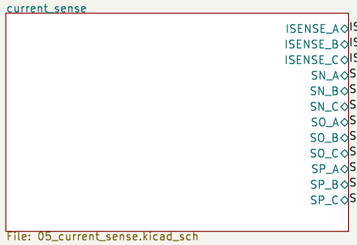
File: 03_bridge.kicad_sch

gate_drive



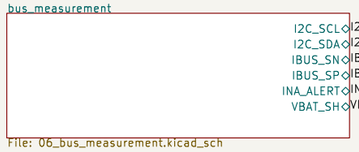
File: 04_gate_drive.kicad_sch

current_sense



File: 05_current_sense.kicad_sch

bus_measurement



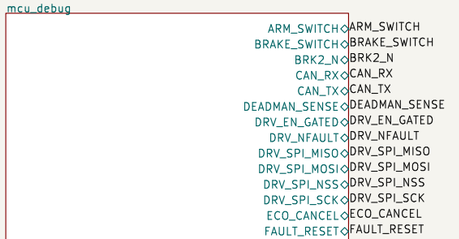
File: 06_bus_measurement.kicad_sch

temperature



File: 07_temperature.kicad_sch

mcu_debug



File: 08_mcu_debug.kicad_sch

hall_inputs



File: 09_hall_inputs.kicad_sch

driver_controls



File: 10_driver_controls.kicad_sch

can_phy



File: 11_can_phy.kicad_sch

aux_rails

File: 12_aux_rails.kicad_sch

Generated by tools/gen_kicad_schematic.py — edit the generator, not the files
Pinouts marked UNVERIFIED must be checked against primary datasheets
Functional schematic draft: ratings UNRESOLVED until motor/battery measured

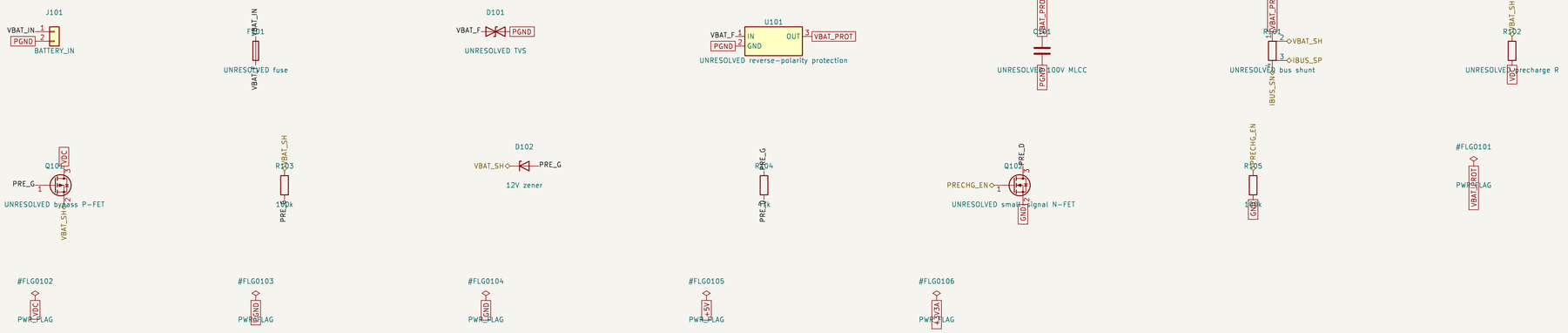
SWORT Aerospace / SEM team

Sheet: /
File: semmc.kicad_sch

Title: SEM motor controller — DRAFT — NOT FOR FABRICATION

Size: A3	Date: 2026-10-15	**Rev: D0.2-draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/13

Precharge: the bypass FET is OFF at power-up (R pull-down on PRECHG_EN, MCU pins high-Z in reset).
Firmware closes it only after V_DC >= 0.95 x V_BAT (VBUS_SENSE vs INA228 VBUS).
The vehicle e-stop isolates the battery upstream of J (vehicle hardware, not this board).



Pin_check UNVERIFIED = pinout not checked against a primary datasheet
UNRESOLVED = depends on motor/battery/harness/rules inputs

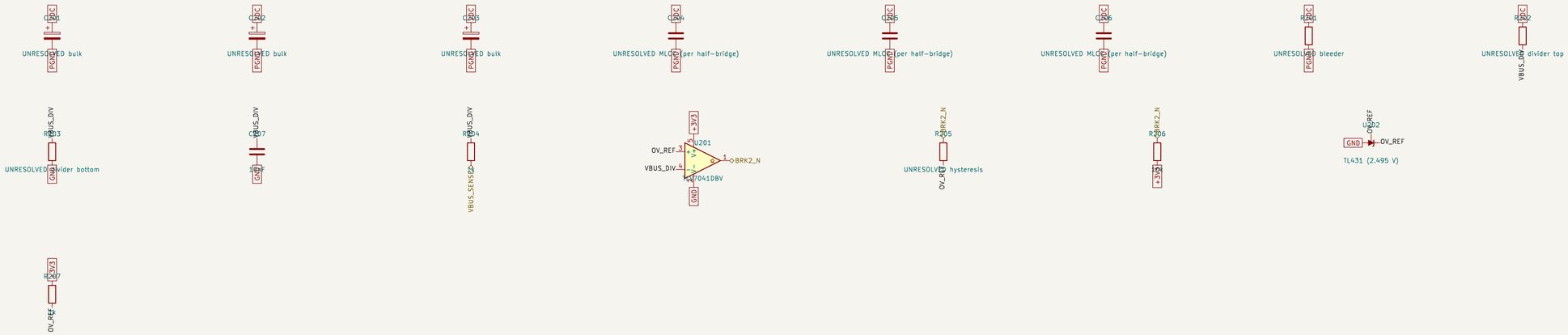
SWORT Aerospace / SEM team

Sheet: /power_input/
File: 01_power_input.kicad_sch

Title: Power input, protection, bus shunt, precharge — DRAFT, NOT FOR FABRI

Size: A2	Date: 2026-10-15	**Rev: D0.2-draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/13

Over-voltage trip is independent of firmware: comparator -> BRK2_N -> TIM1_BKIN2 (outputs off in hardware).



Pin_check UNVERIFIED = pinout not checked against a primary datasheet
UNRESOLVED = depends on motor/battery/harness/rules inputs

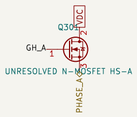
SWORT Aerospace / SEM team

Sheet: /dc_link/
File: 02_dc_link.kicad_sch

Title: DC link, bus-voltage sense, independent over-voltage trip — DRAFT, NO

Size: A2	Date: 2026-10-15	**Rev: D0.2-draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/13

All MOSFET, shunt and connector ratings are UNRESOLVED until motor and battery ratings are measured.











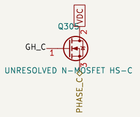






















Pin_check UNVERIFIED = pinout not checked against a primary datasheet
UNRESOLVED = depends on motor/battery/harness/rules inputs

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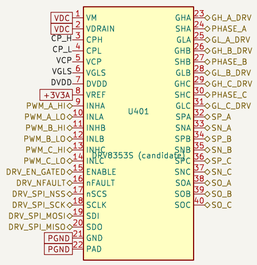
Sheet: /bridge/
File: 03_bridge.kicad_sch

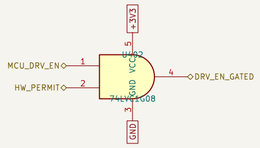
Title: Six-switch bridge and low-side shunts — DRAFT, NOT FOR FABRICATION

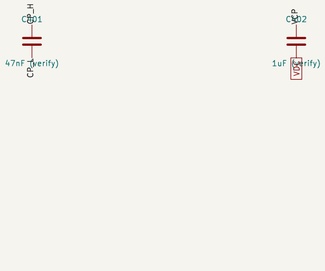
Size: A2	Date: 2026-10-15	**Rev: D0.2-draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/13

DRV8353S pin NAMES and NUMBERS are PLACEHOLDERS (primary datasheet not readable here): renumber from TI SLVSDY6x before any footprint/layout.

Gate-driver ENABLE = MCU_DRV_EN AND HW_PERMIT (safety loop + watchdog): firmware alone cannot enable the driver.







Pin_check UNVERIFIED = pinout not checked against a primary datasheet

UNRESOLVED = depends on motor/battery/harness/rules inputs

SWORT Aerospace / SEM team

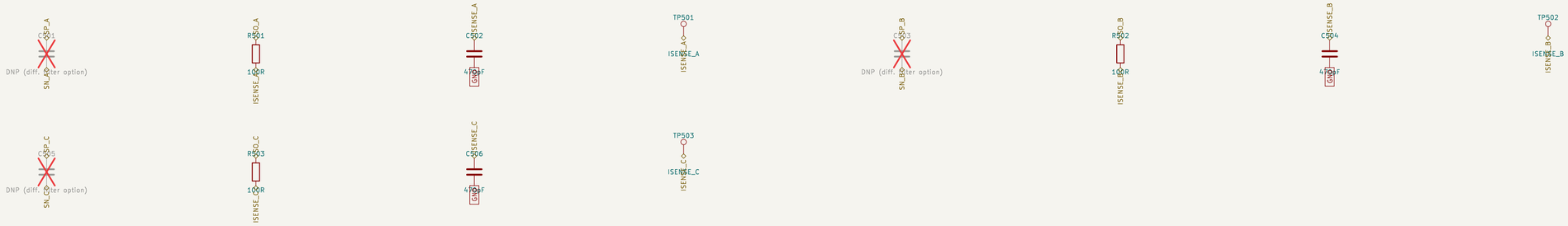
Sheet: /gate_drive/

File: 04_gate_drive.kicad_sch

Title: Gate driver, enable chain, startup states — DRAFT, NOT FOR FABRICATIO

Size: A2	Date: 2026-10-15	**Rev: D0.2-draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/13

PROPOSED, not substantiated: phase currents to ADC1, ADC2 and ADC3 started by the same TIM1 trigger. Simultaneity still needs the trigger configuration, ADC synchronization, acquisition windows and a bench measurement (hardware/timing/timing_budget.md, bring-up stage 4).



Pin_check UNVERIFIED = pinout not checked against a primary datasheet
UNRESOLVED = depends on motor/battery/harness/rules inputs

SWORT Aerospace / SEM team

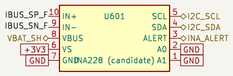
Sheet: /current_sense/
File: 05_current_sense.kicad_sch

Title: Phase-current sensing (CSA outputs to three ADCs) — DRAFT, NOT FOR

Size: A2	Date: 2026-10-15	**Rev: D0.2-draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/13

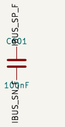
Energy is integrated from this meter, never from the phase shunts.

Its VBUS input is on the battery side of the precharge, so the precharge check is VBUS_SENSE (link) versus INA228 VBUS (battery).

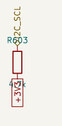


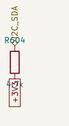














Pin_check UNVERIFIED = pinout not checked against a primary datasheet

UNRESOLVED = depends on motor/battery/harness/rules inputs

SWORT Aerospace / SEM team

Sheet: /bus_measurement/

File: 06_bus_measurement.kicad_sch

Title: DC-bus energy meter (INA228-class) — DRAFT, NOT FOR FABRICATION

Size: A2	Date: 2026-10-15	**Rev: D0.2-draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 7/13

A missing or implausible temperature derates, it does not fault (firmware MC_DR_TEMP_UNKNOWN).















Pin_check UNVERIFIED = pinout not checked against a primary datasheet
UNRESOLVED = depends on motor/battery/harness/rules inputs

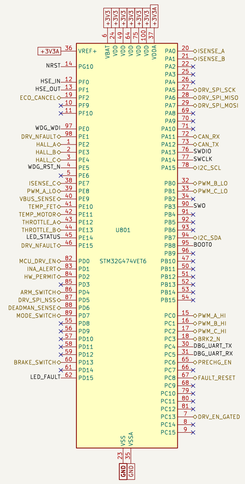
SWORT Aerospace / SEM team

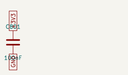
Sheet: /temperature/
File: 07_temperature.kicad_sch

Title: Power-stage and motor temperature — DRAFT, NOT FOR FABRICATION

Size: A2	Date: 2026-10-15	**Rev: D0.2-draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 8/13

Unused MCU pins are no-connect in this draft; pin functions come from hardware/interfaces/pin_allocation_STM32G474VBCETx.csv.







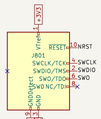








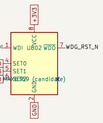






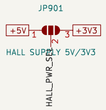





UB02
WDI
VCC
SET0
SET1
WDO
WDG_WDI
WDG_RST_N
GND
+3V3
GND
NC
NC
NC

Hall output stage (open collector vs push-pull) and supply are unknown.





























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UNRESOLVED = depends on motor/battery/harness/rules inputs

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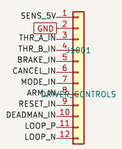
Sheet: /hall_inputs/
File: 09_hall_inputs.kicad_sch

Title: Hall sensor inputs — DRAFT, NOT FOR FABRICATION

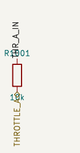
Size: A2	Date: 2026-10-15	**Rev: D0.2-draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 10/13

HW_PERMIT is high only while the vehicle safety loop (e-stop + dead-man) is energised and the watchdog is not asserting.

Brake switch: normally closed preferred (fails safe); the firmware polarity is configurable.





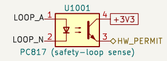














Pin_check UNVERIFIED = pinout not checked against a primary datasheet

UNRESOLVED = depends on motor/battery/harness/rules inputs

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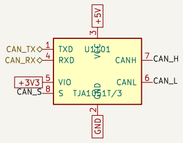
Sheet: /driver_controls/

File: 10_driver_controls.kicad_sch

Title: Wired driver controls and safety-loop sense — DRAFT, NOT FOR FABRIC

Size: A2	Date: 2026-10-15	**Rev: D0.2-draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 11/13

500 kbit/s classic CAN, IDs 0x300–0x33F (docs/INTERFACE_CONTROL.md). Bit timing and termination must match the vehicle bus.

























Pin_check UNVERIFIED = pinout not checked against a primary datasheet
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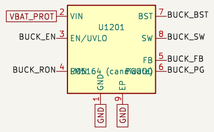
SWORT Aerospace / SEM team

Sheet: /can_phy/
File: 11_can_phy.kicad_sch

Title: CAN physical layer — DRAFT, NOT FOR FABRICATION

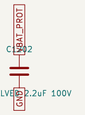
Size: A2	Date: 2026-10-15	**Rev: D0.2-draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 12/13

Logic rails run from VBAT_PROT (before precharge), so the MCU is alive while the link precharges.





UNRESOLVED





UNRESOLVED UVLO top



UNRESOLVED UVLO bottom



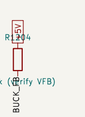
UNRESOLVED RON



2.2nF (verify)

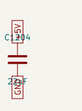


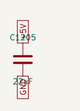
UNRESOLVED Inductor



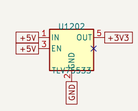
31.6k (verify VFB)

















PGND–GND star



5V



GND



BUCK_PG

Pin_check UNVERIFIED = pinout not checked against a primary datasheet
UNRESOLVED = depends on motor/battery/harness/rules inputs

SWORT Aerospace / SEM team

Sheet: /aux_rails/
File: 12_aux_rails.kicad_sch

Title: Auxiliary rails and ground star point — DRAFT, NOT FOR FABRICATION

Size: A2	Date: 2026-10-15	**Rev: D0.2-draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 13/13